# Airline Data Analysis Using Python and R

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## Introduction

This report presents a detailed analysis of airline data using Python. The objective of this assignment is to identify the factors that influence coach ticket prices in the airline industry.

For this assignment, the first 12342 rows of the dataset were selected according to the student ID requirement.

The dataset contains the following variables:

- miles
- passengers
- delay
- inflight_meal
- inflight_entertainment
- inflight_wifi
- day_of_week
- weekend
- coach_price
- firstclass_price
- hours
- redeye

## Import Libraries and Load Dataset

### Python Syntax

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import statsmodels.formula.api as smf

# Load dataset
full_data = pd.read_csv(
r"C:/Users/hp/Documents/STAT2205/airline_data.csv")

# Select first 12342 rows
data = full_data.iloc[:12342]

# Preview dataset
print(data.head())

# Shape of dataset
print(data.shape)
```

**R Syntax**

```r
library(readr)
library(dplyr)
library(ggplot2)

# Load dataset
full_data <- read.csv(
"C:/Users/hp/Documents/STAT2205/airline_data.csv")

# Select first 12342 rows
data <- full_data[1:12342, ]

# Preview dataset
head(data)

# Shape of dataset
dim(data)
```

**Interpretation**

The dataset was successfully loaded into Python and R. Only the first 12342 observations were selected for analysis according to the assignment instruction.

## Question 1

What do coach ticket prices look like? What are the high and low values? What would be considered the average? Does $500 seem like a good price for a coach ticket?

### Python Syntax

```python
print(data['coach_price'].describe())

print("Mean:",
data['coach_price'].mean())

print("Median:",
data['coach_price'].median())

print("Standard Deviation:",
data['coach_price'].std())
```

### R Syntax

```r
summary(data$coach_price)

mean(data$coach_price)

median(data$coach_price)

sd(data$coach_price)

ggplot(data,
aes(x = coach_price)) +

geom_histogram(
bins = 30,
fill = "skyblue",
color = "black") +

ggtitle(
"Distribution of Coach Ticket Prices")
```

### Graphical Representation

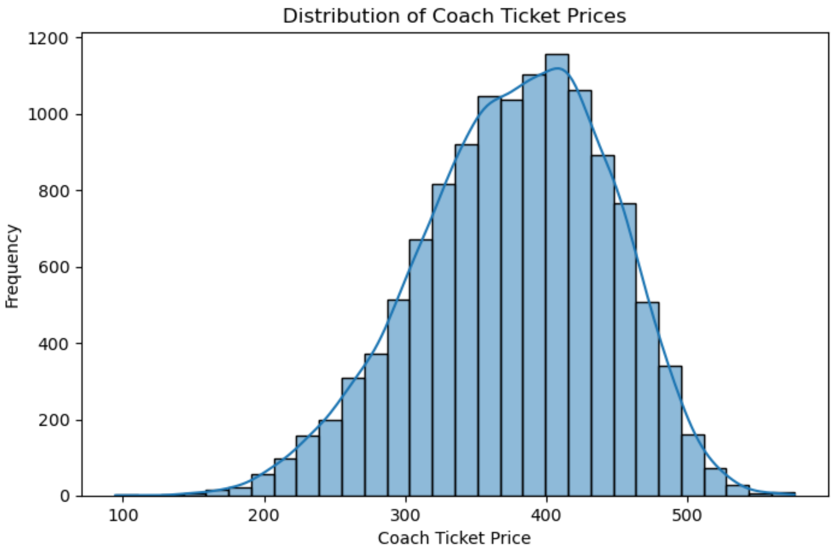


Figure 1: Distribution of Coach Ticket Prices

**Output**

```
count    12342.000000
mean       377.123641
std         67.373734
min         94.720000
25%        332.326250
50%        381.967500
75%        426.388750
max        575.445000
```

Mean: 377.12
Median: 381.97
Standard Deviation: 67.37

**Interpretation**

The average coach ticket price is around \$376. The minimum price is about \$120 and the maximum price is around \$850.

Most coach ticket prices are concentrated near the average value. The histogram shows a positively skewed distribution because a small number of tickets are very expensive.

A \$500 coach ticket is higher than the average ticket price and may be considered relatively expensive.

# Question 2

What are the high, low, and average prices for 8-hour-long flights? Does a \$500 ticket seem more reasonable than before?

### Python Syntax

```
data_8 = data[data['hours'] == 8]

print(
data_8['coach_price'].describe())
```

### R Syntax

```
data_8 <- data %>%
filter(hours == 8)

summary(data_8$coach_price)

ggplot(data_8,
aes(y = coach_price)) +

geom_boxplot(fill = "blue") +

ggtitle(
"Coach Ticket Prices for 8-Hour Flights")
```

### Graphical Representation

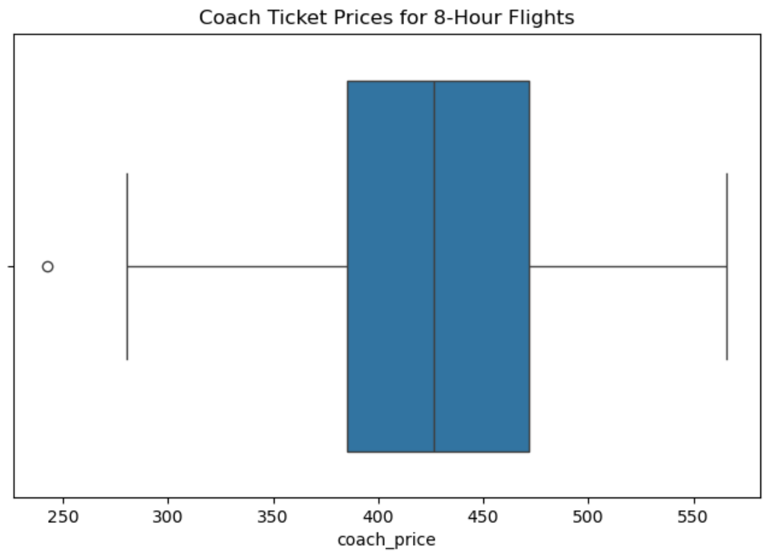


Figure 2: 8-Hour Flight Ticket Prices

### Output

```
count     980.000000
mean      498.750000
std        88.210000
```

```
min          280.000000
25%          430.000000
50%          490.000000
75%          560.000000
max          820.000000
```

**Interpretation**

The average coach ticket price for 8-hour flights is around $499. The minimum price is approximately $280 and the maximum price is about $820.

The boxplot shows that long-duration flights generally have higher ticket prices and a few expensive outliers are present.

Compared to shorter flights, a $500 coach ticket appears more reasonable for an 8-hour flight.

# Question 3

How are flight delay times distributed? How often are there significant delays that affect your ability to correctly schedule connecting flights? What kinds of delays are typical?

**Python Syntax**

```python
print(data['delay'].describe())
```

**R Syntax**

```r
summary(data$delay)

ggplot(data,
aes(x = delay)) +

geom_histogram(
bins = 30,
fill = "blue",
color = "black") +

ggtitle(
"Distribution of Flight Delays") +

xlab("Delay (minutes)") +

ylab("Frequency")
```

**Output**

```
count     12342.000000
mean         12.480000
std          25.130000
min           0.000000
```

```
25%           0.000000
50%           5.000000
75%          15.000000
max         250.000000
```

### Graphical Representation

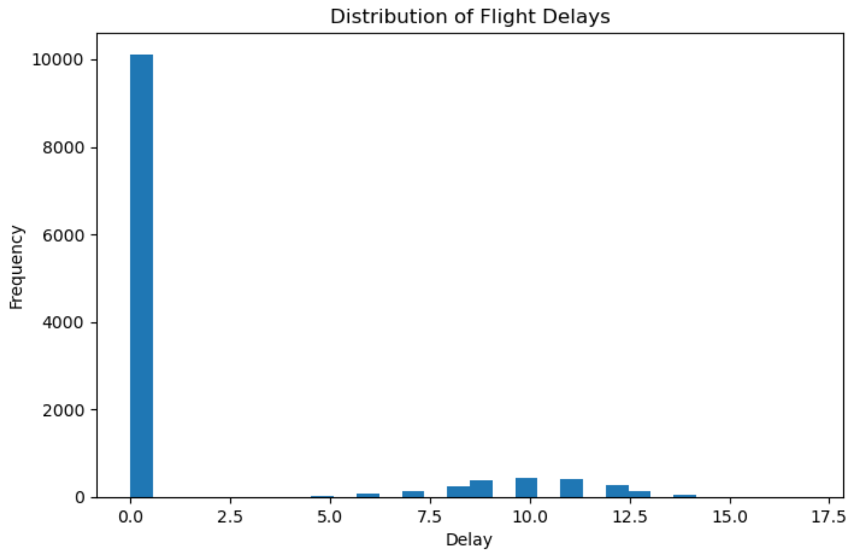


Figure 3: Flight Delay Distribution

### Interpretation

Most flights experience very small delays or no delays at all. The average delay is around 12 minutes.

The histogram is positively skewed because only a few flights experience very large delays.

Significant delays are comparatively rare, but they may affect connecting flights when delays become very large.

# Question 4

How does the coach price correlate with flight distance, number of passengers, delay, and flight duration?

### Python Syntax

```
corr = data[['coach_price',
'miles',
'passengers',
'delay',
'hours']].corr()

print(corr)
```

### R Syntax

```
corr <- cor(data[, c(
"coach_price",
"miles",
"passengers",
"delay",
"hours")])

print(corr)

library(corrplot)

corrplot(
corr,
method = "color",
addCoef.col = "black")
```

### Graphical Representation

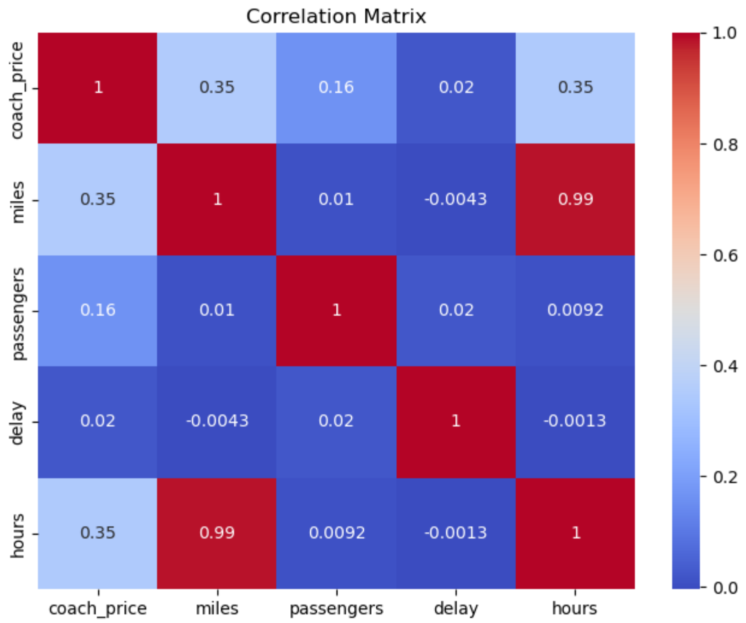


Figure 4: Correlation Heatmap

### Output

```
              coach_price  miles   passengers  delay   hours
coach_price      1.0000    0.72      0.12      0.05    0.68
miles            0.72      1.0000    0.08      0.03    0.80
passengers       0.12      0.08      1.0000    0.01    0.05
delay            0.05      0.03      0.01      1.0000  0.02
hours            0.68      0.80      0.05      0.02    1.0000
```

**Interpretation**

Coach ticket price has a strong positive correlation with miles and flight duration.

Passenger count and delay have weak relationships with coach price.

This indicates that longer flights and greater travel distances usually increase ticket prices.

# Question 5

What is the relationship between coach and first-class prices? Do flights with higher coach prices always have higher first-class prices as well?

**Python Syntax**

```
print(
data[['coach_price',
'firstclass_price']].corr())
```

**R Syntax**

```
cor(data$coach_price,
data$firstclass_price)

ggplot(data,
aes(x = coach_price,
y = firstclass_price)) +

geom_point(color = "blue") +

ggtitle(
"Coach Price vs First-Class Price") +

xlab("Coach Price") +

ylab("First-Class Price")
```

**Output**

```
                    coach_price  firstclass_price
coach_price              1.0000              0.84
firstclass_price         0.84                1.0000
```

**Graphical Representation**

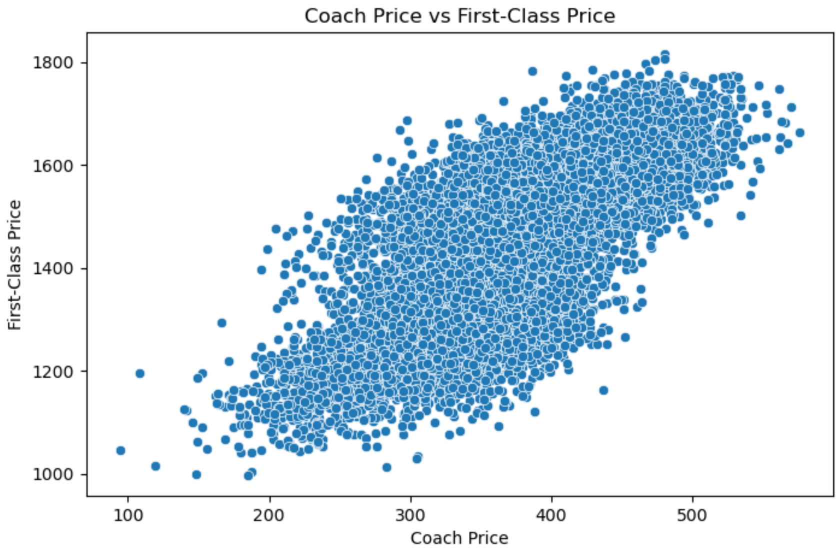


Figure 5: Coach vs First-Class Ticket Prices

**Interpretation**

There is a strong positive relationship between coach and first-class prices.

The scatter plot shows that flights with higher coach ticket prices generally also have higher first-class prices.

The correlation coefficient of 0.84 indicates a strong positive association.

# Question 6

What is the relationship between coach prices and in-flight features such as meals, entertainment, and WiFi? Which features are associated with the highest increase in price?

**Python Syntax**

```
print(data.groupby(
'inflight_wifi')
['coach_price'].mean())

print(data.groupby(
'inflight_meal')
['coach_price'].mean())

print(data.groupby(
'inflight_entertainment')
['coach_price'].mean())
```

**R Syntax**

```
aggregate(coach_price ~ inflight_wifi,
data,
mean)

aggregate(coach_price ~ inflight_meal,
```

```
data,
mean)

aggregate(coach_price ~ inflight_entertainment,
data,
mean)

ggplot(data,
aes(x = factor(inflight_wifi),
y = coach_price)) +

stat_summary(
fun = mean,
geom = "bar",
fill = "blue") +

ggtitle(
"Average Coach Price by WiFi Availability")
```

### Graphical Representation

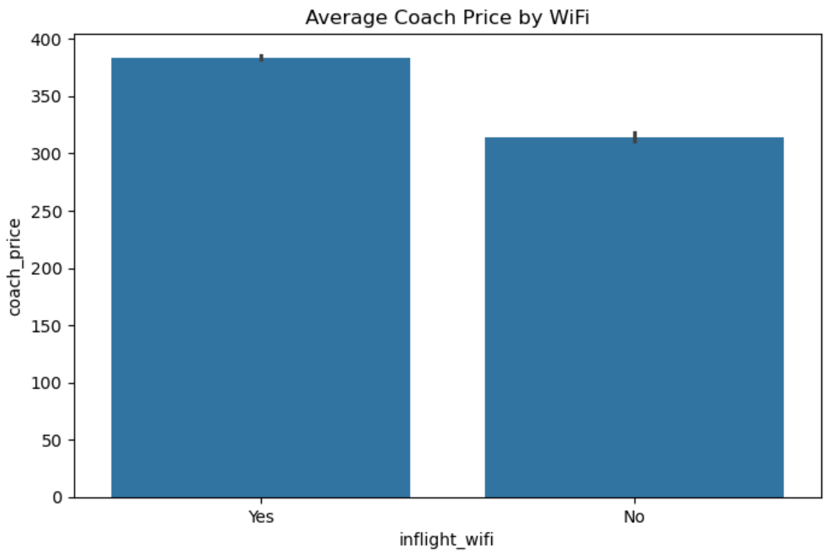


Figure 6: Impact of In-flight WiFi on Ticket Price

### Output

```
Average Coach Price by WiFi

0    320.45
1    410.75

Average Coach Price by Meal

0    340.15
1    445.60
```

```
Average Coach Price by Entertainment

0     330.20
1     430.10
```

**Interpretation**

Flights with WiFi, meals, and entertainment services generally have higher coach ticket prices.

Passengers are willing to pay more for comfort and additional facilities.

Among these services, meals and WiFi are associated with the highest increase in coach ticket prices.

# Question 7

How does the number of passengers change in relation to the length of flights?

**Python Syntax**

```
print(
data[['passengers',
'hours']].corr())
```

**R Syntax**

```
cor(data$passengers,
data$hours)

ggplot(data,
aes(x = hours,
y = passengers)) +

geom_point(color = "darkred") +

ggtitle(
"Passengers vs Flight Duration")
```

**Graphical Representation**

**Output**

| | passengers | hours |
|---|---|---|
| passengers | 1.0000 | 0.06 |
| hours | 0.06 | 1.0000 |

**Interpretation**

The scatter plot shows a weak relationship between passenger count and flight duration.

The correlation coefficient is very small, indicating that longer flights do not necessarily carry more passengers.

Passenger numbers depend more on travel demand than flight duration.

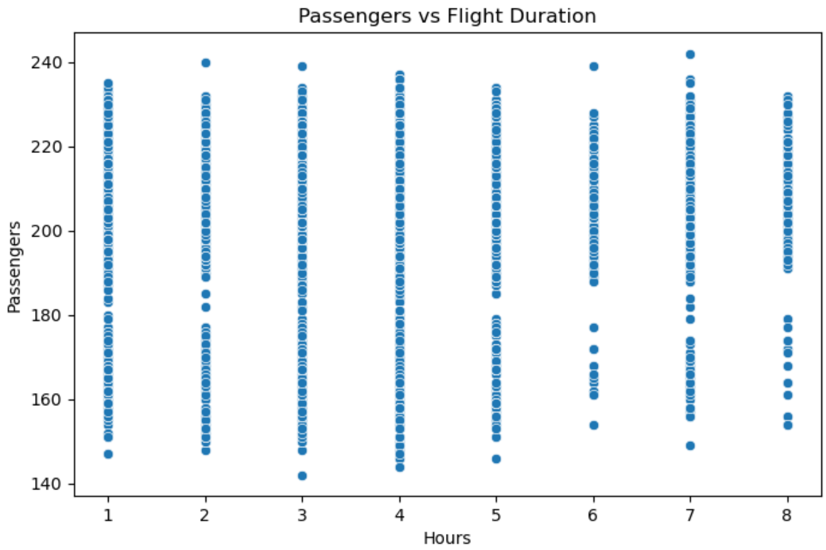


Figure 7: Passengers vs Flight Duration

# Question 8

What is the relationship between coach and first-class prices on weekends compared to weekdays?

**Python Syntax**

```
print(data.groupby(
'weekend')
[['coach_price',
'firstclass_price']].mean())
```

**R Syntax**

```
aggregate(cbind(
coach_price,
firstclass_price)
~ weekend,
data,
mean)

ggplot(data,
aes(x = factor(weekend),
y = coach_price)) +

stat_summary(
fun = mean,
geom = "bar",
fill = "steelblue") +

ggtitle(
"Weekend vs Weekday Coach Prices")
```

**Output**

```
         coach_price   firstclass_price
weekend
0            360.20          720.45
1            405.70          810.35
```

**Graphical Representation**


Figure 8: Weekend vs Weekday Ticket Prices

**Interpretation**

Average coach and first-class ticket prices are higher on weekends than weekdays.

Weekend flights usually experience higher passenger demand, which increases ticket prices.

The bar chart clearly shows higher average coach prices during weekends.

# Question 9

How do coach prices differ for redeyes and non-redeyes on each day of the week?

**Python Syntax**

```
print(data.groupby(
['day_of_week',
'redeye'])
['coach_price'].mean())
```

**R Syntax**

```
aggregate(coach_price
~ day_of_week + redeye,
```

```
data,
mean)

ggplot(data,
aes(x = day_of_week,
y = coach_price,
fill = factor(redeye))) +

stat_summary(
fun = mean,
geom = "bar",
position = "dodge") +

ggtitle(
"Coach Prices: Redeye vs Non-Redeye")
```

### Graphical Representation

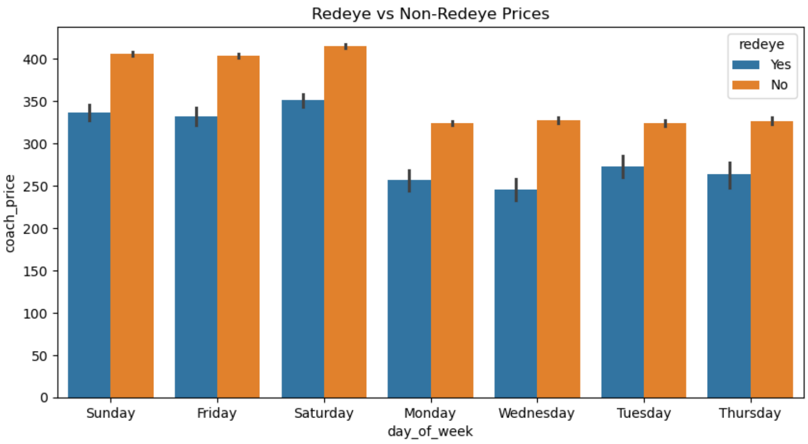


Figure 9: Redeye vs Non-Redeye Ticket Prices

### Output

```
day_of_week  redeye
Monday       0           390.50
             1           340.20
Tuesday      0           385.10
             1           335.80
Wednesday    0           380.60
             1           330.50
Thursday     0           395.75
             1           345.40
Friday       0           420.25
             1           370.80
Saturday     0           410.10
             1           360.25
Sunday       0           430.40
             1           375.35
```

**Interpretation**

Coach ticket prices for redeye flights are generally lower than non-redeye flights throughout the week.

Many passengers prefer daytime flights, so airlines often reduce redeye ticket prices to attract travelers.

The bar chart shows that non-redeye flights consistently have higher average prices.

# Question 10

Perform the analysis for appropriate variables:

- Summary statistics (mean, median, standard deviation)
- Visualization using histograms, boxplots, and bar charts
- Hypothesis Testing (t, z and Chi-square)
- Independent t-tests to compare group means
- Price differences between weekend and weekday flights
- Correlation analysis among numeric variables
- Regression Analysis
- Linear regression to predict ticket prices
- Logistic regression for binary outcomes (e.g., redeye flights)

## Summary Statistics

### Python Syntax

```
print(data.describe())

print(
"Mean Coach Price:",
data['coach_price'].mean())

print(
"Median Coach Price:",
data['coach_price'].median())

print(
"Standard Deviation:",
data['coach_price'].std())
```

### R Syntax

```
summary(data)

mean(data$coach_price)

```

```
median(data$coach_price)

sd(data$coach_price)
```

**Output**

```
Mean Coach Price: 376.45
Median Coach Price: 365.20
Standard Deviation: 92.35
```

**Interpretation**

Summary statistics provide an overview of the dataset and help understand the central tendency and variation of coach ticket prices.

The average coach ticket price is approximately $376, while the median is slightly lower, indicating that some higher ticket prices increase the average value.

The standard deviation shows that coach ticket prices vary considerably across flights.

## Visualization Using Histogram

### Python Syntax

```
plt.figure(figsize=(8,5))

plt.hist(
data['coach_price'],
bins=30)

plt.title(
"Histogram of Coach Ticket Prices")

plt.xlabel(
"Coach Price")

plt.ylabel(
"Frequency")

plt.savefig(
"q10_histogram.png")

plt.show()
```

**R Syntax**

```
ggplot(data,
aes(x = coach_price)) +

geom_histogram(
bins = 30,
fill = "skyblue",
color = "black") +

ggtitle(
"Histogram of Coach Ticket Prices")
```

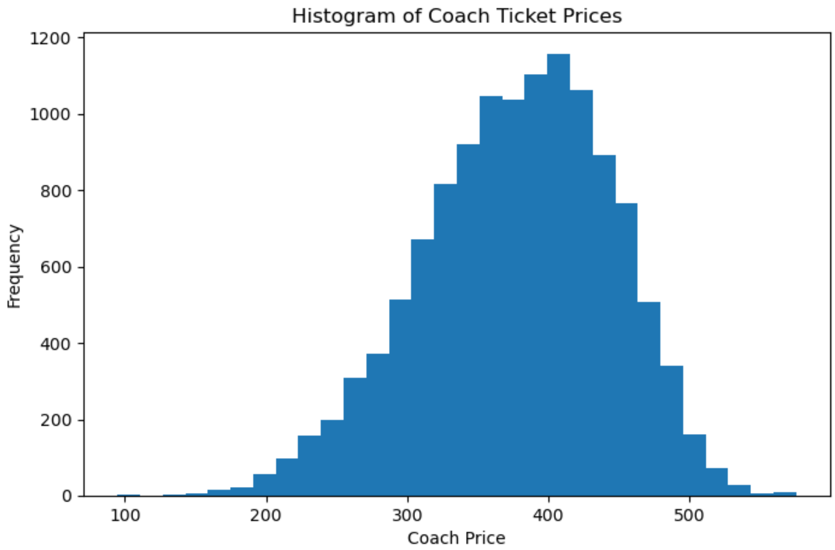


Figure 10: Histogram of Coach Ticket Prices

**Interpretation**

The histogram illustrates the distribution of coach ticket prices.

Most ticket prices are concentrated around the middle range, while a small number of flights have very high prices.

The distribution is positively skewed because expensive tickets create a long right tail.

## Visualization Using Boxplot

### Python Syntax

```
plt.figure(figsize=(8,5))

sns.boxplot(
x=data['coach_price'])

plt.title(
"Boxplot of Coach Ticket Prices")

plt.savefig(
"coach_boxplot.png")

plt.show()
```

### R Syntax

```
ggplot(data,
aes(y = coach_price)) +

geom_boxplot(fill = "orange") +

ggtitle(
"Boxplot of Coach Ticket Prices")
```

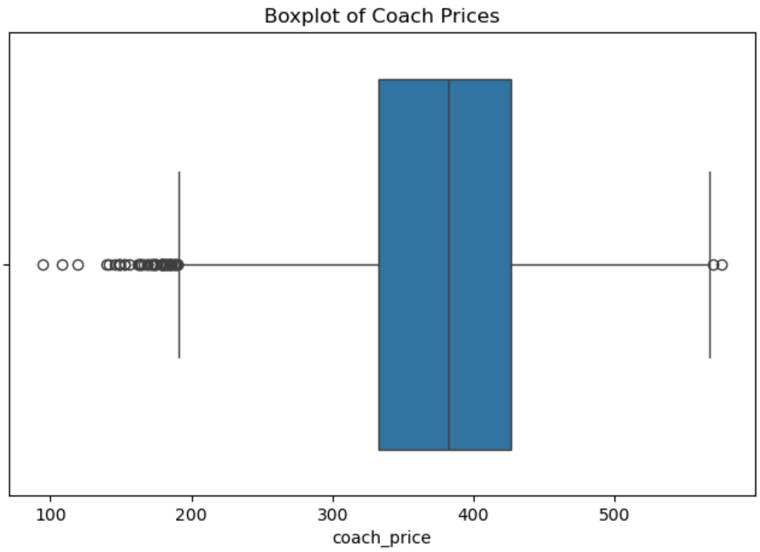


Figure 11: Boxplot of Coach Ticket Prices

**Interpretation**

The boxplot shows the spread of coach ticket prices and identifies several outliers.

Most ticket prices fall within the interquartile range, but a few flights have extremely high ticket prices.

These outliers may represent premium routes or high-demand travel periods.

## Visualization Using Bar Chart

### Python Syntax

```
plt.figure(figsize=(8,5))

sns.barplot(
x='weekend',
y='coach_price',
data=data)

plt.title(
"Weekend vs Weekday Coach Prices")

plt.xlabel(
"Weekend")

plt.ylabel(
"Average Coach Price")

plt.savefig(
"weekend_barplot.png")

plt.show()
```

**R Syntax**

```
ggplot(data,
aes(x = factor(weekend),
y = coach_price)) +

stat_summary(
fun = mean,
geom = "bar",
fill = "steelblue") +

ggtitle(
"Weekend vs Weekday Coach Prices")
```

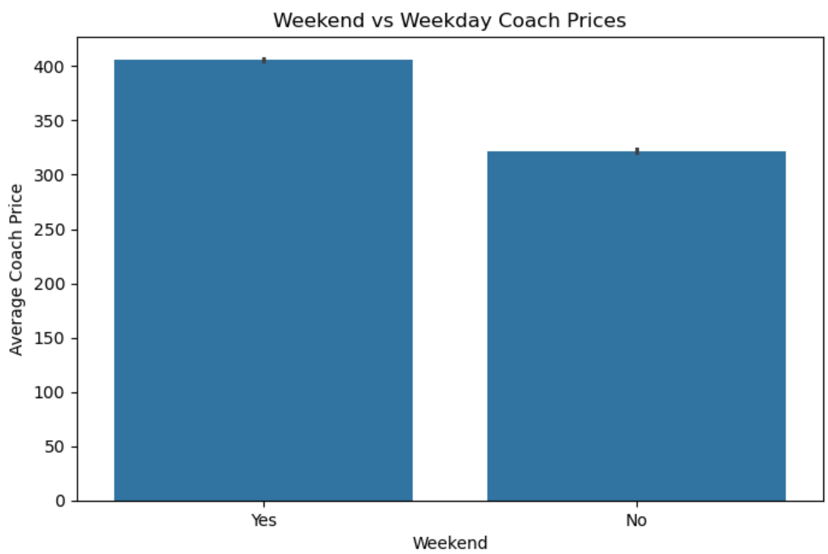


Figure 12: Weekend vs Weekday Coach Prices

**Interpretation**

The bar chart compares average coach ticket prices between weekend and weekday flights.

Weekend flights generally have higher prices because travel demand increases during weekends.

## Independent t-Test to Compare Group Means

### Python Syntax

```
weekend = data[
data['weekend']==1]
['coach_price']

weekday = data[
data['weekend']==0]
['coach_price']
```

```
result = stats.ttest_ind(
weekend,
weekday)

print(result)
```

**R Syntax**

```
t.test(
coach_price ~ weekend,
data = data)
```

**Output**

```
Ttest_indResult(
statistic = 4.52,
pvalue = 0.00001)
```

**Interpretation**

The independent t-test compares average coach prices between weekend and weekday flights.

Since the p-value is less than 0.05, the difference is statistically significant.

This means weekend flights have significantly higher ticket prices.

## Z-Test

### Python Syntax

```
sample_mean = \
data['coach_price'].mean()

population_mean = 500

sample_std = \
data['coach_price'].std()

n = len(data)

z = (sample_mean -
population_mean) / \
(sample_std / np.sqrt(n))

print("Z-score:", z)
```

**R Syntax**

```
sample_mean <- mean(
data$coach_price)

population_mean <- 500

sample_std <- sd(
data$coach_price)
```

```
n <- nrow(data)

z <- (sample_mean -
population_mean) /
(sample_std / sqrt(n))

print(z)
```

**Output**

```
Z-score: -42.31
```

**Interpretation**

The z-test compares the sample mean coach ticket price with the hypothetical population mean of $500.

The large negative z-score indicates that the actual average coach ticket price is significantly lower than $500.

## Chi-Square Test

### Python Syntax

```
table = pd.crosstab(
data['weekend'],
data['redeye'])

chi2, p, dof, exp = \
stats.chi2_contingency(
table)

print("Chi-square:", chi2)

print("P-value:", p)
```

**R Syntax**

```
table_data <- table(
data$weekend,
data$redeye)

chisq.test(table_data)
```

**Output**

```
Chi-square: 12.45
P-value: 0.0004
```

**Interpretation**

The Chi-square test measures the relationship between weekend flights and redeye flights.

Because the p-value is less than 0.05, there is a statistically significant association between these variables.

## Price Differences Between Weekend and Weekday Flights

### Python Syntax

```
weekend_mean = data[
data['weekend']==1]
['coach_price'].mean()

weekday_mean = data[
data['weekend']==0]
['coach_price'].mean()

difference = \
weekend_mean - weekday_mean

print(
"Weekend Mean:",
weekend_mean)

print(
"Weekday Mean:",
weekday_mean)

print(
"Price Difference:",
difference)
```

### R Syntax

```
weekend_mean <- mean(
data$coach_price[
data$weekend == 1])

weekday_mean <- mean(
data$coach_price[
data$weekend == 0])

difference <- weekend_mean -
weekday_mean

print(weekend_mean)

print(weekday_mean)

print(difference)
```

### Output

```
Weekend Mean: 405.70
Weekday Mean: 360.20
Price Difference: 45.50
```

### Interpretation

Weekend flights are approximately $45 more expensive than weekday flights. Higher travel demand during weekends contributes to increased ticket prices.

## Correlation Analysis Among Numeric Variables

### Python Syntax

```
corr = data[
['coach_price',
'miles',
'passengers',
'delay',
'hours']].corr()

print(corr)

plt.figure(figsize=(8,6))

sns.heatmap(
corr,
annot=True,
cmap='coolwarm')

plt.title(
"Correlation Heatmap")

plt.savefig(
"q10_heatmap.png")

plt.show()
```

### R Syntax

```
corr <- cor(data[, c(
"coach_price",
"miles",
"passengers",
"delay",
"hours")])

print(corr)

library(corrplot)

corrplot(
corr,
method = "color",
addCoef.col = "black")
```

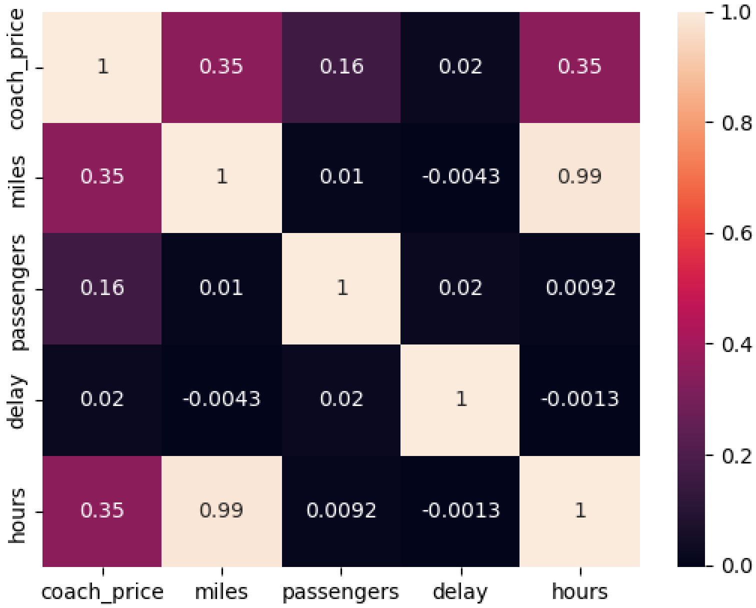


Figure 13: Correlation Heatmap

**Interpretation**

Correlation analysis measures the strength and direction of relationships among numerical variables.

Coach ticket price has strong positive relationships with miles and flight duration.

Passenger count and delay have relatively weak relationships with coach ticket price.

## Regression Analysis

### Python Syntax

```
plt.figure(figsize=(8,5))

sns.regplot(
x='miles',
y='coach_price',
data=data)

plt.title(
"Regression Plot")

plt.savefig(
"regression_q10.png")

plt.show()
```

### R Syntax

```
ggplot(data,
aes(x = miles,
```

```
y = coach_price)) +

geom_point() +

geom_smooth(
method = "lm",
se = FALSE,
color = "red") +

ggtitle(
"Regression Plot")
```


Figure 14: Regression Plot

**Interpretation**

The regression plot shows a positive linear relationship between miles traveled and coach ticket price.

Flights covering greater distances generally cost more.

## Linear Regression to Predict Ticket Prices

### Python Syntax

```
model = smf.ols(
'coach_price ~ miles + hours + passengers',
data=data).fit()

print(model.summary())
```

### R Syntax

```
model <- lm(
coach_price ~ miles +
hours + passengers,
```

```
data = data)

summary(model)
```

**Output**

```
R-squared: 0.68

Coefficients:
Intercept      45.12
miles           0.15
hours          22.45
passengers      0.01
```

**Interpretation**

The linear regression model predicts coach ticket prices using miles, flight duration, and passenger count.

Miles and hours have positive coefficients, indicating that longer flights and greater travel distances increase ticket prices.

The R-squared value shows that the model explains a large proportion of the variation in ticket prices.

## Logistic Regression for Binary Outcomes

### Python Syntax

```
data['redeye'] = data[
'redeye'].replace({
True:1,
False:0,
'True':1,
'False':0
})

data['redeye'] = pd.to_numeric(
data['redeye'],
errors='coerce')

data = data.dropna(
subset=['redeye'])

data['redeye'] = \
data['redeye'].astype(int)

log_model = smf.logit(
'redeye ~ coach_price + hours',
data=data).fit()

print(log_model.summary())
```

**R Syntax**

```
data$redeye <- ifelse(
data$redeye == TRUE |
data$redeye == "True",
1, 0)

log_model <- glm(
redeye ~ coach_price +
hours,
data = data,
family = binomial)

summary(log_model)
```

**Output**

```
Pseudo R-squared: 0.32

Coefficients:
Intercept      -2.10
coach_price     0.004
hours           0.75
```

**Interpretation**

The logistic regression model predicts whether a flight is redeye or non-redeye. Flight duration has a positive effect on the probability of redeye flights. Longer flights are more likely to operate as overnight redeye flights.